

DSAA3073: Theories in Data Science

Tutorial 4A: Diagnose Error and Stabilize an Inverse

150-minute core tutorial - 15-minute break - 60-minute protected buffer

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Explorative projection

LESSON BACKBONE

When a fitted predictor changes sharply across samples, how can we separate the source of prediction error, locate the unstable directions, and modify the inverse problem without pretending that noise has disappeared?

Tutorial 3B gave us a statistical view of model choice: empirical fit must be read together with a complexity certificate. This tutorial examines a complementary mechanism. At one fixed input, repeated fits can fluctuate because the data used to estimate them fluctuate. In a linear inverse problem, small singular directions reveal exactly where such perturbations can be amplified.

Chapter 1 already established the compact SVD $A = U_r \Sigma_r V_r^T$, the relations $Av_j = \sigma_j u_j$, and the positive-definite-Hessian certificate for strict convexity. We will retrieve those results rather than prove them again. The new work is to turn singular directions into a stability diagnosis and then into a regularized inverse.

We will first separate squared prediction error into systematic displacement, sample-to-sample spread, and fresh observation noise. We will then distinguish an absent singular direction from a small but nonzero one. Finally, we will change the optimization problem by charging for solution size and calculate how that charge damps each direction.

The three core blocks take **50, 45, and 55 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-50	Derive and diagnose pointwise squared-loss bias-variance	50 min core
50-65	Scheduled break	15 min
65-110	Classify inverse problems and audit singular-direction amplification	45 min core
110-165	Formulate and solve the finite-dimensional Tikhonov problem	55 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

One test input, two kinds of randomness

Fix a test input x_0 . A learning algorithm receives a random training sample S and returns a fitted function $\hat{f}_S$. Its prediction at the fixed point, $\hat{f}_{S(x_0)}$, is random because a different sample generally produces a different fit. A fresh response at the same point is

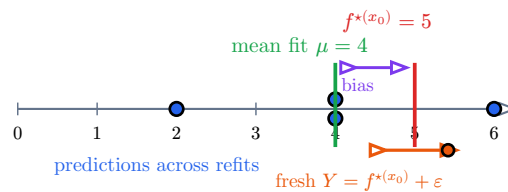
$$Y = f^*(x_0) + \varepsilon, \quad (1)$$

where $f^*(x_0) = \mathbb{E}[Y \mid x_0]$ and ε is fresh test noise. Throughout this block we assume

$$\mathbb{E}[\varepsilon \mid x_0] = 0, \quad \text{Var}(\varepsilon \mid x_0) = \sigma^2(x_0), \quad (2)$$

and the fresh ε is independent of the training sample S . We also use squared prediction loss. These are assumptions, not consequences of the notation.

Imagine four independent repetitions of the whole training procedure. At x_0 , they produce the values 2, 4, 4, 6. Their average is 4, while the regression value is $f^*(x_0) = 5$.



Blue marks are four training repetitions. The independent orange interval and dot show a fresh response draw around the fixed red $f^*(x_0)$.

Figure 1: At a fixed input, the distance between the green and red markers is systematic displacement; the blue spread is training-sample variation. Fresh response noise is a separate random source.

Before any formula is named, mark the roles:

- x_0 and $f^*(x_0)$ are fixed in the pointwise calculation;
- S is random and controls $\hat{f}_{S(x_0)}$;
- ε is a fresh random quantity independent of S ;
- the two expectations must not be silently merged before their roles are clear.

Observable action. Point to the random object represented by the blue spread, then point to the different random object represented by the orange interval. The correct answer is: blue changes when the training sample S is replaced and the model is refitted; orange changes when a fresh ε is drawn after the fit. The red regression value is fixed in both repetitions.

Definition: Squared-loss bias-variance decomposition

Let

$$\mu(x_0) = \mathbb{E}_S[\hat{f}_{S(x_0)}]. \quad (3)$$

Under the fixed-input, squared-loss, zero-conditional-mean, and freshness assumptions above,

$$\mathbb{E}_{S,\varepsilon}[(Y - \hat{f}_{S(x_0)})^2] = \underbrace{(\mu(x_0) - f^*(x_0))^2}_{\text{squared bias}} + \underbrace{\text{Var}_S(\hat{f}_{S(x_0)})}_{\text{variance across training samples}} + \underbrace{\sigma^2(x_0)}_{\text{fresh-noise variance}}. \quad (4)$$

The word **pointwise** matters: x_0 stays fixed. Integrating the identity over a distribution of test inputs is another valid step, but it is not hidden in this statement.

Why the cross terms vanish

Write $\hat{f} = \hat{f}_{S(x_0)}$, $\mu = \mathbb{E}_S[\hat{f}]$, and $f^* = f^*(x_0)$ for this calculation. Add and subtract μ :

$$Y - \hat{f} = (f^* - \mu) + (\mu - \hat{f}) + \varepsilon. \quad (5)$$

After squaring and taking expectations, three square terms and three cross terms appear. The first training-sample cross term is

$$\mathbb{E}_S[(f^* - \mu)(\mu - \hat{f})] = (f^* - \mu)(\mu - \mathbb{E}_S[\hat{f}]) = 0. \quad (6)$$

For a term containing fresh noise, condition on the training sample:

$$\mathbb{E}_{(S,\varepsilon)}[(\mu - \hat{f})\varepsilon] = \mathbb{E}_S[(\mu - \hat{f})\mathbb{E}[\varepsilon \mid S, x_0]] = 0. \quad (7)$$

Freshness and zero conditional mean give $\mathbb{E}[\varepsilon \mid S, x_0] = 0$. The remaining cross term, $\mathbb{E}[(f^* - \mu)\varepsilon]$, vanishes for the same reason. The square terms are

$$(f^* - \mu)^2, \quad \mathbb{E}_S[(\hat{f} - \mu)^2] = \text{Var}_S(\hat{f}), \quad \mathbb{E}[\varepsilon^2 \mid x_0] = \sigma^2(x_0). \quad (8)$$

That proves the identity. If the test noise were correlated with the training sample, or had nonzero conditional mean, the displayed decomposition would need additional terms.

Worked example: Four refits at one test input

For the predictions 2, 4, 4, 6, the mean prediction is $\mu = 4$. With $f^*(x_0) = 5$ and $\sigma^2(x_0) = 3$,

$$\text{squared bias} = (4 - 5)^2 = 1, \quad (9)$$

$$\text{variance} = \frac{(2 - 4)^2 + (4 - 4)^2 + (4 - 4)^2 + (6 - 4)^2}{4} = 2. \quad (10)$$

The expected squared prediction error is therefore $1 + 2 + 3 = 6$. This is a population-style calculation over the declared four equally weighted training-sample outcomes; replacing it by an unbiased sample-variance estimator would answer a different estimation question.

IN-CLASS CHECK

Prompt: A second method has mean prediction 5 and variance 4 at the same x_0 . The fresh-noise variance is still 3. Compare its MSE with the first method and diagnose the difference.

Hint: Compute squared bias, training-sample variance, and noise separately.

Answer:

The second method has squared bias $(5 - 5)^2 = 0$, variance 4, and noise 3, so its MSE is 7. The first method had MSE 6. Removing one unit of squared bias did not compensate for adding two units of training-sample variance. The noise term is unchanged because the observation mechanism is unchanged.

IN-CLASS CHECK

Prompt: Which assumption justifies $\mathbb{E}[(\mu - \hat{f})\varepsilon] = 0$? Is the statement $\mathbb{E}[\varepsilon] = 0$ by itself sufficient?

Hint: Condition first on the random object that produced $\hat{f}$.

Answer:

We use $\mathbb{E}[\varepsilon | S, x_0] = 0$, supplied here by fresh zero-mean noise independent of the training sample. An unconditional equality $\mathbb{E}[\varepsilon] = 0$ alone is not sufficient: ε could still be correlated with $f_{S(x_0)}$, leaving a nonzero cross term.

A related decomposition with different objects

Tutorial 3 used approximation and estimation error to compare a learned hypothesis with the best hypothesis in a class and with the population optimum. That is a class-level risk comparison. The identity just proved instead fixes x_0 , assumes squared loss, averages a prediction across repeated training samples, and includes fresh response noise.

Both viewpoints warn that flexibility and data dependence must be balanced. They are not interchangeable formulas. In particular, there is no universal law saying that prediction variance halves whenever the sample size doubles; that behavior depends on the estimator, design, and data distribution.

The decomposition tells us **what** changed across refits. It has not yet told us **which mathematical direction** made the fit sensitive. To find that direction, we now read fitting as an inverse problem.

When an inverse exists but should not be trusted

Consider a forward map A that turns an unknown vector f into an observation $y = Af$. The inverse task starts with y and asks for f . Before computing an inverse matrix, we should ask three separate questions about the declared admissible data and solution spaces.

Definition: Hadamard well-posedness

An inverse problem is **well posed** on its declared admissible inputs when:

1. a solution exists for every admissible observation;
2. that solution is unique in the modeled solution class; and
3. the solution depends continuously on the observation.

If any one condition fails, the problem is **ill posed**. Existence, uniqueness, and continuous dependence are three different claims; proving one never silently proves the other two.

For an invertible matrix between fixed finite-dimensional Euclidean spaces, the inverse is continuous. Yet its operator norm can be enormous. The formal continuity claim then coexists with poor quantitative conditioning: a small measurement error can still produce an unusably large change in the recovered vector.

Two inverse geometries to classify

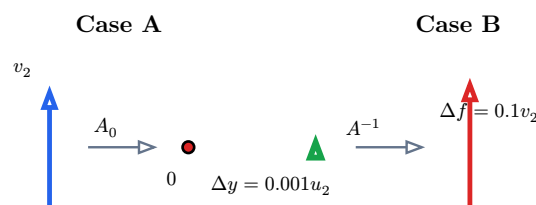


Figure 2: Two raw forward/inverse geometries. Classify them before reading the reveal below.

IN-CLASS CHECK

Prompt: Classify which Hadamard conditions can fail for Case A, represented by $\text{diag}(1, 0)$, and which statement is correct for Case B, represented by $\text{diag}(1, 0.01)$.

Hint: For Case A, test one observation in the range and one outside it. For Case B, compute the inverse multiplier in the displayed second direction.

Answer:

Stop here and commit two diagnoses before turning the page. Your written pair is the observable response; the complete answer and numerical audit begin under the explicit reveal on the next page.

Reveal: zero and small are qualitatively different

Question	Case A	Case B
second singular value	0	0.01
inverse multiplier	undefined on that direction	$\frac{1}{0.01} = 100$
diagnosis	check existence and uniqueness	unique but quantitatively sensitive

Answer reveal. For $\text{diag}(1, 0)$, a range observation can have many preimages, while an observation outside the range has none; uniqueness and existence can fail. For $\text{diag}(1, 0.01)$, every observation has a unique solution and the inverse is continuous, but its norm is 100, so the problem is badly conditioned.

For Case A, $A_0 = \text{diag}(1, 0)$. If $y = (2, 0)$, every $f = (2, t)$ is a solution, so uniqueness fails. If $y = (2, 0.01)$ is admitted, there is no exact solution, so existence fails. One zero singular direction can therefore create different failures for different observations.

For Case B, let $A = \text{diag}(1, 0.01)$ and $y = (2, 0.03)$. The unique exact solution is $f = (2, 3)$. Perturb only the second observation coordinate:

$$\Delta y = (0, 0.001) \implies \Delta f = A^{-1} \Delta y = (0, 0.1). \quad (11)$$

Thus $\frac{\|\Delta f\|_2}{\|\Delta y\|_2} = 100$. The inverse is still continuous in this finite-dimensional case. Calling it discontinuous would be wrong; the correct diagnosis is severe quantitative sensitivity.

Definition: Singular-value conditioning of a linear inverse

Retrieve the Chapter 1 compact SVD $A = U_r \Sigma_r V_r^T$, so the active singular-vector relations are

$$A v_j = \sigma_j u_j, \quad (12)$$

with $\sigma_j > 0$ and orthonormal right and left singular vectors. On an active direction with $\sigma_j > 0$, exact inversion recovers the coordinate by

$$\langle f, v_j \rangle = \frac{\langle y, u_j \rangle}{\sigma_j}. \quad (13)$$

A small σ_j therefore amplifies an observation perturbation in the u_j direction by $\frac{1}{\sigma_j}$. For an invertible matrix, the Euclidean condition number is

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}. \quad (14)$$

The condition number is a worst-case **relative** sensitivity measure. The reciprocal $\frac{1}{\sigma_{\min}}$ is an **absolute** inverse amplification bound. Keeping these roles separate prevents unit and scale errors.

IN-CLASS CHECK

Prompt: Let $A = \text{diag}(4, 0.02)$. A perturbation of size 0.006 lies entirely in the second left-singular direction. Compute the recovered perturbation size and $\kappa_2(A)$.

Hint: Divide by the matching singular value; use the ratio of largest to smallest for the condition number.

Answer:

The recovered perturbation size is $\frac{0.006}{0.02} = 0.3$. The condition number is $\frac{4}{0.02} = 200$. The first number is the actual amplification for the declared perturbation; the second is the worst-case relative conditioning ratio.

The historical Week 4 sheet contains a diagonal numerical display whose matrix, right-hand side, stated solution, and perturbation coordinate do not satisfy the displayed multiplication. We do not repair it by changing one number and keeping the conclusion. The two maps above are a fresh forward-map calculation, and every factor follows from $Av_j = \sigma_j u_j$.

Small singular directions provide a mechanism for refit instability: a small change in observations can cause a large coefficient change and, depending on the test input, a large prediction change. This does not mean that a condition number equals the statistical variance from the first block. It identifies a route by which observation variation can be amplified.

Discarding the direction would remove its instability, but also all of its signal. We want a modified inverse that keeps the direction while reducing its gain.

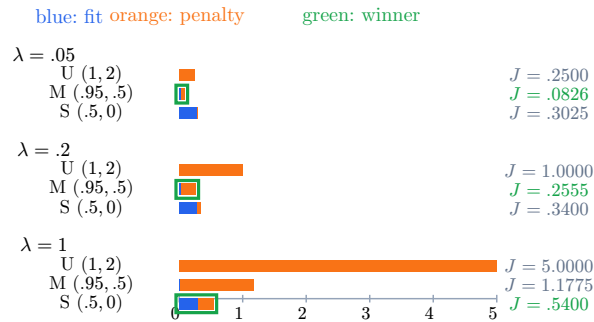
Pay for solution size

Return to $A = \text{diag}(1, 0.1)$ and observation $y = (1, 0.2)$. Exact inversion gives $f = (1, 2)$, but the second coordinate has inverse gain 10. Three candidate solutions display the tradeoff before we name it.

candidate f	$\ Af - y\ ^2$	$\ f\ ^2$
$(1, 2)$	0	5
$(0.95, 0.5)$	0.025	1.1525
$(0.5, 0)$	0.29	0.25

Table 1: The two ingredients are visible, but the winning candidate has not yet been revealed.

Observable action before the reveal. For each of $\lambda = .05, .2, 1$, compute $J_\lambda = \text{data fit} + \lambda \times \text{solution size}$ for the three rows. Record only the winning row. Then uncover the stacked bars below and audit both the segment lengths and the green winner outline.



Minimizer path among the three candidates: $M \rightarrow M \rightarrow S$ as $\lambda : .05 \rightarrow .2 \rightarrow 1$.

Figure 3: Every bar uses the same linear J_λ scale. Blue is $\|Af - y\|^2$; orange is $\lambda\|f\|^2$. Only the true winner at each λ receives a green outline.

Reveal answer. The middle candidate wins for $\lambda = .05$ and $.2$; the small candidate wins for $\lambda = 1$. In particular, the J_1 winner is $(.5, 0)$, not the middle row. This comparison is only among three candidates; the actual minimizer is found over the whole solution space.

Definition: Tikhonov regularization

Let $\mathcal{H}$ and $\mathcal{K}$ be Hilbert spaces, let $A : \mathcal{H} \rightarrow \mathcal{K}$ be a declared forward operator, let $y \in \mathcal{K}$ be observed data, and let $\lambda > 0$. In the zero-prior, squared-norm form used in this course, Tikhonov regularization solves

$$f_\lambda \in \arg \min_{f \in \mathcal{H}} \{\|Af - y\|^2 + \lambda\|f\|^2\}. \quad (15)$$

The first term is data misfit. The second charges for solution size. The parameter λ controls their relative weight; it does not remove noise from y .

The registered objective is $\|Af - y\|^2 + \lambda\|f\|^2$. A prior-centered variant replaces $\|f\|^2$ by $\|f - f_0\|^2$, but the zero-prior form is the interface that Tutorial 4B will reuse. In infinite-

dimensional or constrained problems, existence of a minimizer requires assumptions; the direct calculations below are finite dimensional.

The finite-dimensional minimizer

Take $A \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, $f \in \mathbb{R}^d$, Euclidean norms, and $\lambda > 0$. Differentiating

$$J_{\lambda(f)} = \|Af - y\|_2^2 + \lambda\|f\|_2^2 \quad (16)$$

gives

$$\nabla J_{\lambda(f)} = 2A^T(Af - y) + 2\lambda f. \quad (17)$$

Every minimizer therefore satisfies the **normal equations**

$$(A^T A + \lambda I)f_\lambda = A^T y. \quad (18)$$

The matrix $A^T A + \lambda I$ will be shown positive definite below and is therefore invertible, so the normal equations have the explicit solution $f_\lambda = (A^T A + \lambda I)^{-1} A^T y$.

For every nonzero z ,

$$z^T(A^T A + \lambda I)z = \|Az\|_2^2 + \lambda\|z\|_2^2 > 0. \quad (19)$$

Hence the Hessian $2(A^T A + \lambda I)$ is positive definite. By the Chapter 1 second-order certificate, the objective is strictly convex and therefore has at most one minimizer. The displayed closed form supplies existence, so together the two statements give exactly one minimizer. Keeping these jobs separate matters: positive curvature proves uniqueness, not existence by itself. This remains a finite-dimensional linear result, not a general nonlinear or infinite-dimensional theorem.

Worked example: Solve the two-direction problem

For $A = \text{diag}(1, 0.1)$ and $y = (1, 0.2)$,

$$f_\lambda = \left((1 + \lambda)^{-1}, \frac{0.02}{0.01 + \lambda} \right). \quad (20)$$

At $\lambda = .05$ this is approximately $(0.9524, 0.3333)$; at $\lambda = .2$ it is $(0.8333, 0.0952)$; at $\lambda = 1$ it is $(0.5, 0.0198)$. The optimizer is not restricted to the three candidates in the earlier table. As λ grows, both coordinates move toward zero, but the unstable second coordinate is damped much more sharply.

Read one singular direction at a time

Reuse the Chapter 1 compact SVD. Let $r = \text{rank}(A)$ and write its positive singular pairs as $Av_j = \sigma_j u_j$ for $j = 1, \dots, r$, where $\sigma_j > 0$. Decompose

$$f = \sum_{j=1}^r c_j v_j + f_0, \quad f_0 \in \ker(A). \quad (21)$$

Resolving the normal equations in these positive right-singular directions gives

$$f_\lambda = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \lambda} \langle y, u_j \rangle v_j, \quad P_{\ker(A)} f_\lambda = 0. \quad (22)$$

Any component of y orthogonal to $\text{range}(A)$ contributes a constant to the least-squares misfit and therefore no solution coefficient. The null-space component is separate: because the prior is zero and $\lambda > 0$, minimizing the penalty forces $f_0 = 0$.

Three facts are now visible.

1. If $\sigma_j > 0$, unregularized inversion uses gain $\frac{1}{\sigma_j}$, while the regularized data-to-solution gain is $\frac{\sigma_j}{\sigma_j^2 + \lambda}$.
2. Relative to the unregularized coefficient in that direction, the multiplier is $\frac{\sigma_j^2}{\sigma_j^2 + \lambda}$, a number strictly between zero and one.
3. Directions in $\ker(A)$ do not belong to the positive-pair sum. The data contain no information about them, and the Euclidean zero-prior penalty sets their separate null-space component to zero; it does not reconstruct missing information.

For fixed $\lambda > 0$, the data-to-solution map is continuous, with

$$\|f_{\lambda(y+\Delta y)} - f_{\lambda(y)}\|_2 \leq \max_{\sigma \geq 0} \frac{\sigma}{\sigma^2 + \lambda} \|\Delta y\|_2 = \frac{1}{2\sqrt{\lambda}} \|\Delta y\|_2. \quad (23)$$

The maximum occurs at $\sigma = \sqrt{\lambda}$. The bound makes the stability gain explicit, while also showing the price of taking λ too small.

IN-CLASS CHECK

Prompt: For a positive singular pair with $\sigma = 0.1$, compare the unregularized inverse gain with the Tikhonov gain at $\lambda = 0.05$. What fraction of the unregularized coefficient remains?

Hint: Use $\frac{1}{\sigma}$, $\frac{\sigma}{\sigma^2 + \lambda}$, and $\frac{\sigma^2}{\sigma^2 + \lambda}$.

Answer:

Unregularized inversion has gain 10. The regularized gain is $\frac{0.1}{0.01+0.05} = \frac{3}{5} \approx 1.667$. The remaining fraction is $\frac{0.01}{0.01} = \frac{6}{10} = \frac{3}{5}$. The observation direction is retained, but its influence on the solution is strongly damped.

IN-CLASS CHECK

Prompt: Why does $\lambda > 0$ guarantee a unique finite-dimensional minimizer even when A has a nontrivial null space?

Hint: Test the quadratic form of the Hessian on a nonzero vector.

Answer:

The Hessian is $2(A^T A + \lambda I)$. For every nonzero z , $z^T(A^T A + \lambda I)z = \|Az\|_2^2 + \lambda\|z\|_2^2 > 0$. Thus Chapter 1's PD-Hessian certificate makes the objective strictly convex, hence it has at most one minimizer; the normal-equation solution proves that one exists. A null-space vector may make the first term zero, but the positive penalty term remains.

What the zero-penalty limit actually selects

The phrase “ λ tends to zero, so we recover the inverse” needs a rank qualification. In finite-dimensional linear least squares with the Euclidean penalty and exact minimizers,

$$f_\lambda = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \lambda} \langle y, u_j \rangle v_j, \quad r = \text{rank}(A), \quad P_{\ker(A)} f_\lambda = 0. \quad (24)$$

As $\lambda \rightarrow 0$ through positive values, each of the r positive-pair coordinates tends to $\frac{\langle y, u_j \rangle}{\sigma_j}$, while every null-space coordinate remains zero. Thus

$$f_\lambda \rightarrow \sum_{j=1}^r \frac{\langle y, u_j \rangle}{\sigma_j} v_j = A^\dagger y. \quad (25)$$

Therefore the limit is the Moore-Penrose minimum-norm least-squares solution. If the system is consistent, it is the minimum-norm exact solution; if it is inconsistent, it is the minimum-norm vector among least-squares minimizers. Under full column rank, this reduces to the unique ordinary least-squares solution.

This conclusion uses the finite-dimensional linear spectral form, a Euclidean zero-prior penalty, and $\lambda \rightarrow 0$ through positive values. It is not an unconditional claim about arbitrary Tikhonov problems. The minimum-norm branch agrees with Mohri, Section 11.3.1, equation (11.11), while the positive- λ quadratic form is the ridge form in Section 11.3.2, equations (11.12)–(11.14). Tutorial 4B will specialize the calculation to regression; no regression formula is needed yet.

The general convergence statement stays schematic

For a general operator and admissible set, fixed positive regularization can stabilize the data-to-solution relation only under suitable analytic assumptions. Recovering an appropriate unregularized solution as noise vanishes requires a compatible co-limit: the noise level tends to zero, the regularization parameter tends to zero, and neither the noise nor optimization error overwhelms the penalty. A representative schedule has $\frac{\delta^2}{\lambda} \rightarrow 0$ together with $\lambda \rightarrow 0$, plus the relevant existence, compactness or closedness, range, and approximation assumptions.

The core claim is deliberately schematic. It says which quantities must be coordinated and why a fixed positive λ is not itself an exact-recovery limit. It does not import a full infinite-dimensional convergence theorem into the finite-dimensional calculation.

IN-CLASS CHECK

Prompt: A student proposes $\lambda_\delta = \delta^3$ because both quantities then tend to zero. Does this satisfy the representative requirement $\frac{\delta^2}{\lambda_\delta} \rightarrow 0$?

Hint: Simplify the ratio before interpreting it.

Answer:

No. The ratio is $\frac{\delta^2}{\lambda_\delta} = \frac{\delta^2}{\delta^3} = \frac{1}{\delta}$, which diverges. Merely sending both quantities to zero is not enough; their relative rate matters. For example, $\lambda_\delta = \delta$ gives $\frac{\delta^2}{\lambda_\delta} = \delta \rightarrow 0$.

Standard language to carry forward

Term or notation	Meaning in this tutorial	Recall cue
squared-loss bias-variance decomposition	Pointwise MSE equals squared bias plus variance across training samples plus fresh-noise variance under the declared assumptions.	$\text{MSE} = \text{bias}^2 + \text{variance} + \sigma^2$
Hadamard well-posedness	Existence, uniqueness, and continuous dependence on admissible data are three separate requirements.	test all three
singular-value conditioning	Inversion divides an active observation coordinate by its singular value.	$\frac{1}{\sigma_j}$
Tikhonov regularization	Minimize squared data misfit plus a positive weighted squared solution norm.	$\ Af - y\ ^2 + \lambda\ f\ ^2$

Reconstruct the Tutorial 4B entry knowledge

Tutorial 4B draws on two layers. The table below is the exact nine-item handoff produced by Tutorial 4A. In addition, Tutorial 4B may call back the Chapter 1 compact SVD and PD-Hessian certificate already retrieved in this tutorial; those two Core foundations are not newly exported by Tutorial 4A.

Available item	Productive recall
Hilbert-space background	Use inner products and squared norms to read the objective.
linear algebra	Use transposes, positive definiteness, and singular coordinates.
structural risk minimization	Read a fit-complexity compromise without claiming identical guarantees.
empirical risk minimization	Recognize unpenalized data fit as the baseline optimization.
bias-variance decomposition	Reconstruct the fixed-input identity with both random sources.
well-posedness	Audit existence, uniqueness, and continuous dependence separately.
spectral conditioning	Compute inverse amplification from a singular value.
Tikhonov regularization	Derive and interpret the finite-dimensional normal equations.
$\ Af - y\ ^2 + \lambda\ f\ ^2$	Decode the data-misfit term, the penalty, and the role of $\lambda > 0$.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct: (i) the three bias-variance-noise assumptions, (ii) the three Hadamard criteria, and (iii) the Tikhonov objective with every object's role. Then explain in one sentence why $\lambda > 0$ helps a small singular direction.

Hint: Organize the answer as randomness, inverse problem, then modified objective.

Answer:

(i) At fixed x_0 under squared loss, fresh test noise has zero conditional mean and is independent of the random training sample; its conditional variance is $\sigma^2(x_0)$. (ii) A well-posed inverse has existence, uniqueness, and continuous dependence on admissible data. (iii) For $A: \mathcal{H} \rightarrow \mathcal{K}$, observed $y \in \mathcal{K}$, candidate $f \in \mathcal{H}$, and $\lambda > 0$, minimize $\|Af - y\|_2^2 + \lambda \|f\|_2^2$. In singular direction j , the regularized data-to-solution gain is $\frac{\lambda + \frac{\sigma_j^2}{\sigma_j^2}}{\frac{\lambda}{\sigma_j^2}}$ instead of $\frac{\sigma_j^2}{\lambda}$, so a small singular value no longer creates the same inverse amplification.

We now know how to write the stabilizing objective and read its directions. Tutorial 4B will ask what this objective becomes for linear regression, how its prediction bias and variance change direction by direction, and what evidence should govern the decision to regularize. Ridge, parameter-selection methods, and validation remain future topics here.

Optional deep dives

The following material takes **0 class minutes**. It is for post-class or self-study and is not a prerequisite for Tutorial 4B.

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: When does a norm-constrained problem correspond to a Tikhonov-penalized problem?

Answer.

Consider a convex data-misfit function $L(f)$ and the constrained problem

$$\min_f L(f) \quad \text{subject to} \quad \|f\|^2 \leq r^2. \quad (26)$$

If an optimum exists and an appropriate constraint qualification holds, the Karush-Kuhn-Tucker conditions supply a multiplier $\lambda \geq 0$ such that the same optimum minimizes $L(f) + \lambda\|f\|^2$. Conversely, a minimizer f_λ of the penalized problem solves the constrained problem with radius $r = \|f_\lambda\|$.

This is a matched-parameter statement, not a claim that every chosen λ corresponds to every chosen radius, nor that the optimizer is unique without stricter assumptions. Nonconvex problems or failed constraint qualifications can break the clean correspondence. The useful conclusion is conditional: a penalty can act as the Lagrange-multiplier form of a norm budget when the optimization hypotheses support that step.

Post-class or self-study. Not a prerequisite for the core path.

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: What extra structure sits behind the schematic Tikhonov convergence statement?

Answer.

Crepey’s nonlinear inverse-problem formulation makes several obligations visible. Work in a Hilbert space with a nonempty closed convex admissible set, a prior f_0 , a forward operator satisfying the required weak-to-strong compactness property, and exact data lying in the operator’s range. Let noisy observations satisfy $\|y - y_\delta\| \leq \delta$, allow minimization error η_δ , and choose λ_δ so that

$$\lambda_\delta \rightarrow 0, \quad \frac{\delta^2}{\lambda_\delta} \rightarrow 0, \quad \frac{\eta_\delta}{\lambda_\delta} \rightarrow 0. \quad (27)$$

Under those assumptions, sequences of approximate regularized minimizers have convergent subsequences whose limits are solutions closest to the prior; uniqueness of that prior-closest solution upgrades subsequential convergence to convergence of the full sequence.

Quantitative rates require more, such as differentiability and a source condition linking $f - f_0$ to an adjoint range. Those assumptions explain why the core tutorial did not say “let λ go to zero” and stop. The parameter, noise, optimization accuracy, admissible set, forward operator, and target solution must be controlled together.

Post-class or self-study. Not a prerequisite for the core path.

Source note

The historical Week 4 Learning Sheet supplies the official scope and earlier course framing. The squared-loss derivation is reconstructed with its random objects and cross-term conditions made explicit, following the standard bias-variance treatment and Claeskens-Hjort, Section 1.1(ii). Hadamard's three criteria and the qualified nonlinear Tikhonov statements follow Crepey, **Tikhonov Regularization**, Sections 1-2. The finite-dimensional spectral calculation is derived directly from the SVD. The zero-penalty minimum-norm boundary is bound to Mohri et al., Section 11.3.1, equation (11.11), together with the positive-penalty quadratic form in Section 11.3.2, equations (11.12)-(11.14). Shalev-Shwartz and Ben-David, Chapter 13, supports the fit-plus-norm viewpoint while the tutorial keeps statistical guarantees and general inverse-problem convergence within their declared assumptions.

DSAA3073: Theories in Data Science

Tutorial 4B: Read Ridge One Direction at a Time

150-minute core tutorial - 15-minute break - 60-minute protected buffer

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Explorative projection

LESSON BACKBONE

Tikhonov gives us a stabilizing objective. In linear regression, what does λ do in each unstable direction, what prediction cost does it impose, and what evidence justifies regularizing at all?

Tutorial 4A ended with a general squared data-misfit plus squared-norm penalty. Here we specialize that objective to a linear model. The algebra gives a unique coefficient vector for every $\lambda > 0$, but uniqueness is only the beginning: we still need to see which directions move, how their bias and variance change, and whether that change addresses the failure in front of us.

One fixed-design example will carry the argument. Its Gram matrix has one well-determined direction and one weak direction. We will derive ridge, rotate into those directions, and finish with four systems that have the same training error but need different interventions.

The entry knowledge has two layers. Tutorial 4A supplies its exact nine-item handoff. Chapter 1 separately supplies the compact SVD and the PD-Hessian certificate that Tutorial 4A retrieved and that we now use productively for ridge.

The three core blocks take **50, 60, and 40 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-50	Specialize Tikhonov, derive ridge, and certify its boundary cases	50 min core
50-65	Scheduled break	15 min
65-125	Track shrinkage, coefficient bias, and variance by eigendirection	60 min core
125-165	Diagnose the failure before proposing regularization	40 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

From the Tikhonov objective to ridge

Tutorial 4A established exactly nine items that this tutorial may reuse. The four background items are Hilbert-space language, linear algebra, structural risk minimization (SRM), and empirical risk minimization (ERM). The four new mathematical ideas are the squared-loss bias-variance decomposition, Hadamard well-posedness, spectral conditioning, and Tikhonov regularization. The ninth item is the registered objective

$$\|Af - y\|^2 + \lambda\|f\|^2, \quad \lambda > 0. \quad (28)$$

Two Chapter 1 Core foundations remain available in addition to that local nine-item handoff: the compact SVD $X = U_r \Sigma_r V_r^T$ and the rule that a positive-definite Hessian certifies strict convexity. They are prior knowledge, not extra concepts introduced by Tutorial 4A.

We also rely on the course-level prerequisites in probability and calculus. No parameter-selection method is assumed.

IN-CLASS CHECK

Prompt: Before specializing the objective, identify which entry layer justifies each move: differentiating a quadratic, adding a squared norm, certifying uniqueness, and resolving a small singular direction.

Hint: Separate the exact Tutorial 4A handoff from the Chapter 1 Core callbacks and general course prerequisites.

Answer:

Differentiation uses the course calculus prerequisite. The squared solution norm and the positive parameter come from Tikhonov regularization and its registered objective. Concern about a small singular direction comes from spectral conditioning, while Chapter 1's compact SVD resolves that direction. The Chapter 1 PD-Hessian certifies the positive quadratic form into strict convexity and hence uniqueness. General linear algebra still supplies transposes and eigenvalues.

Let $X \in \mathbb{R}^{n \times d}$ be a fixed design matrix, let $y \in \mathbb{R}^n$ be the response vector, and let $\beta \in \mathbb{R}^d$ be the coefficient vector. We work with centered columns and a centered response, so no intercept appears. This is equivalent to fitting an unpenalized intercept and applying the penalty only to the remaining coefficients. Penalizing an intercept would be a different modeling convention.

Replacing A by X and f by β turns the Tikhonov objective into

$$J_{\lambda(\beta)} = \|y - X\beta\|^2 + \lambda\|\beta\|^2, \quad \lambda > 0. \quad (29)$$

Our scaling has no factor $\frac{1}{n}$ in the data-fit term. A source that writes $\frac{\|y - X\beta\|^2}{n} + \lambda'\|\beta\|^2$ uses the same family after the conversion $\lambda = n\lambda'$; one must not mix the two numerical parameter values.

Before revealing the estimator, complete the middle and right stages of this pipeline. Every displayed matrix must have compatible dimensions.

$J_{\lambda(\beta)} = \ y - X\beta\ ^2 + \lambda\ \beta\ ^2$	$\rightarrow$	$\nabla J_{\lambda(\beta)} = ?$	$\rightarrow$	$(X^T X + \lambda I_d)\beta = ?$
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$X^T X \in \mathbb{R}^{d \times d}$, $I_d \in \mathbb{R}^{d \times d}$, and $X^T y \in \mathbb{R}^d$. The smallest eigenvalue of $X^T X + \lambda I_d$ is at least λ .

Table 2: The penalty changes the curvature matrix, not the response vector. Fill the two question marks before continuing.

The missing expressions are $-2X^T y + 2X^T X\beta + 2\lambda\beta$ and $X^T y$. The term λI_d has the same dimension as the Gram matrix $X^T X$.

Definition: Ridge regression

With the centered-design convention above and $\lambda > 0$, **ridge regression** chooses the minimizer of

$$\|y - X\beta\|^2 + \lambda\|\beta\|^2. \quad (30)$$

Under this course's scaling convention, its registered coefficient estimator is

$$\hat{\beta}_\lambda = (X^T X + \lambda I_d)^{-1} X^T y. \quad (31)$$

The squared Euclidean penalty applies to every displayed coefficient; an intercept has already been removed by centering and is therefore not penalized.

Why the minimizer is unique

Expanding the objective and differentiating gives

$$J_{\lambda(\beta)} = y^T y - 2\beta^T X^T y + \beta^T X^T X \beta + \lambda \beta^T \beta, \quad (32)$$

$$\nabla J_{\lambda(\beta)} = -2X^T y + 2(X^T X + \lambda I_d)\beta. \quad (33)$$

Setting the gradient to zero gives the ridge normal equations and the displayed closed form. To show this stationary point is the unique global minimizer, take any nonzero $v \in \mathbb{R}^d$:

$$v^T \nabla^2 J_{\lambda(\beta)} v = 2v^T (X^T X + \lambda I_d) v = 2\|Xv\|^2 + 2\lambda\|v\|^2 > 0. \quad (34)$$

Thus the Hessian is positive definite for $\lambda > 0$, even if X has dependent columns. By the Chapter 1 second-order certificate, the objective is strictly convex and therefore has at most one minimizer. The displayed closed form supplies existence, so together they give a unique coefficient minimizer. This proof does not certify lower prediction risk.

Worked example: A rank-deficient design still has one ridge solution

Suppose

$$X^T X = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}, \quad X^T y = (8, 0)^T, \quad \lambda = 2. \quad (35)$$

The unpenalized normal equations require $4\beta_1 = 8$ and impose no condition on β_2 , so every $(2, t)^T$ is an OLS minimizer. Ridge instead solves

$$\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix} \beta = (8, 0)^T, \quad (36)$$

giving the unique vector $\hat{\beta}_2 = (\frac{4}{3}, 0)^T$. The null-direction coefficient becomes zero because it cannot improve data fit but still pays the penalty.

Ridge changes generic coefficients continuously as λ changes. It does not search over feature subsets, and it does not normally set a coefficient exactly to zero. In the example above, the second coefficient is zero because that direction is an exact null direction of the design, not because ridge performed subset selection. Discrete subset search and squared-norm shrinkage solve different optimization problems.

What happens as the penalty vanishes

Reuse the Chapter 1 compact singular-value decomposition $X = U_r \Sigma_r V_r^T$, where the positive singular values are $\sigma_1, \dots, \sigma_r$. Direct substitution into the ridge formula gives

$$\hat{\beta}_\lambda = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \lambda} \langle u_j, y \rangle v_j. \quad (37)$$

There is no contribution from a null direction. Therefore, as $\lambda \rightarrow 0$,

$$\hat{\beta}_\lambda \rightarrow \sum_{j=1}^r \frac{1}{\sigma_j} \langle u_j, y \rangle v_j = X^+ y. \quad (38)$$

If X has full column rank, $X^+ y = (X^T X)^{-1} X^T y$ is the unique OLS coefficient vector. If X is rank deficient, $X^+ y$ is the Moore-Penrose minimum-norm least-squares solution among all OLS minimizers. Writing an ordinary inverse in the second case would be incorrect. In the other limit, every displayed coefficient tends to zero as $\lambda \rightarrow \infty$.

IN-CLASS CHECK

Prompt: A design has singular values 3 and 0. For $\langle u_1, y \rangle = 6$, compute $\hat{\beta}_\lambda$ in the two right-singular directions and its limit as $\lambda \rightarrow 0$. Why is the limit not an arbitrary OLS solution?

Hint: Use the factor $\frac{\sigma_j}{\sigma_j^2 + \lambda}$ only for positive singular values.

Answer:

The active coefficient is $\frac{18}{\lambda+9}$ and the null-direction coefficient is 0. The limit is $(2, 0)$ in the right-singular basis. OLS permits any null-direction addition, but ridge selects zero there, which is exactly the minimum-norm least-squares choice.

Follow the Gram-matrix directions

Let

$$G = X^T X = Q \operatorname{diag}(\mu_1, \dots, \mu_d) Q^T, \quad (39)$$

where $Q = (q_1, \dots, q_d)$ is orthogonal and every $\mu_j \geq 0$. The numbers μ_j are squared singular values of X . If $z_j = \langle q_j, X^T y \rangle$, then the ridge coefficient in direction q_j is $\frac{z_j}{\mu_j + \lambda}$.

For a positive μ_j , the corresponding full-rank OLS coefficient is $\frac{z_j}{\mu_j}$. Dividing the ridge coefficient by it leaves a number between zero and one. The two curves below are deliberately called **P** and **Q**. Decide which curve belongs to $\mu = 100$ and which belongs to $\mu = 1$; do not turn the page until you have written both assignments and one mathematical reason.

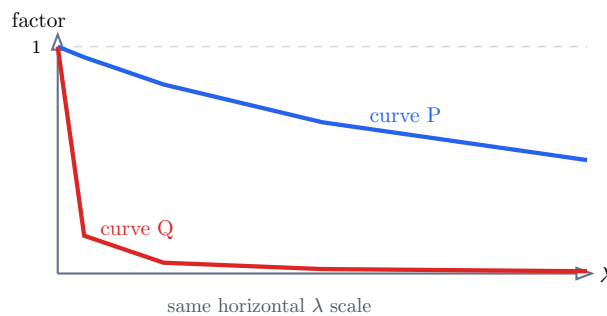


Figure 4: Two unlabeled directional responses to the same horizontal λ scale. Assign $\mu = 100$ and $\mu = 1$ to P and Q before the definition supplies the rule.

Commit before the reveal. Write $P = \dots$, $Q = \dots$, and complete: “At the same positive λ , the direction that keeps the larger fraction is ... because ...”.

Definition: Ridge spectral shrinkage

For a positive eigenvalue μ_j of $X^T X$, ridge multiplies the OLS coefficient in direction q_j by the **shrinkage factor**

$$s_{j(\lambda)} = \frac{\mu_j}{\mu_j + \lambda}. \quad (40)$$

Small- μ_j directions are attenuated most strongly. The inverse factor $\frac{1}{\mu_j + \lambda}$ acts on $z_j = \langle q_j, X^T y \rangle$; it is not the shrinkage factor. When $\mu_j = 0$, ridge assigns zero to that coefficient direction.

Indeed,

$$\hat{\beta}_\lambda = Q \operatorname{diag} \left(\frac{1}{\mu_j + \lambda} \right) Q^T X^T y. \quad (41)$$

For $\mu_j > 0$, insert $\frac{\mu_j}{\mu_j}$ into the j th term:

$$\langle q_j, \hat{\beta}_\lambda \rangle = \underbrace{\frac{\mu_j}{\mu_j + \lambda}}_{s_{j(\lambda)}} \underbrace{\frac{z_j}{\mu_j}}_{\text{OLS coefficient in direction } q_j}. \quad (42)$$

The formula explains the two curves. At a common λ , the fraction kept is close to one when μ_j is large and close to zero when μ_j is small.

Reveal. P is the $\mu = 100$ curve and Q is the $\mu = 1$ curve. Thus the stable direction stays near one while the weak direction collapses first. This is the conclusion to compare with your written commitment, not information that was available from the pre-commitment labels.

IN-CLASS CHECK

Prompt: For $\mu_1 = 100$, $\mu_2 = 1$, and $\lambda = 9$, compute both $\frac{1}{\mu_j + \lambda}$ and $s_{j(\lambda)}$. Which pair should be compared when the question is “what fraction of the OLS coefficient remains”?

Hint: Keep the input to each factor visible.

Answer:

The inverse factors are $\frac{1}{100}$ and $\frac{1}{1}$. The shrinkage factors are $\frac{100}{109} \approx 0.917$ and $\frac{1}{10} = 0.1$. Fractions of the OLS coefficients are given by the shrinkage factors, not by the inverse factors. Thus the weak direction retains only 10 percent while the stable one retains about 91.7 percent.

Put randomness back into the response

To discuss bias and variance, declare the probability model rather than reading them from the deterministic objective. Keep X fixed and suppose

$$y = X\beta^* + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0, \quad \operatorname{Cov}(\varepsilon \mid X) = \sigma^2 I_n. \quad (43)$$

The noise coordinates are uncorrelated, have a common variance, and the linear conditional mean is correctly specified. Under these conditions,

$$\mathbb{E}[\hat{\beta}_\lambda \mid X] = (G + \lambda I)^{-1} G \beta^*, \quad (44)$$

$$\text{Cov}(\hat{\beta}_\lambda \mid X) = \sigma^2 (G + \lambda I)^{-1} G (G + \lambda I)^{-1}. \quad (45)$$

Write $\theta_j = \langle q_j, \beta^* \rangle$. In direction q_j ,

$$\mathbb{E}[\hat{\theta}_{\lambda,j} \mid X] = s_{j(\lambda)} \theta_j, \quad (46)$$

$$\text{bias}_j = -\frac{\lambda}{\mu_j + \lambda} \theta_j, \quad \text{variance}_j = \sigma^2 \frac{\mu_j}{(\mu_j + \lambda)^2}. \quad (47)$$

These are coefficient-space moments. For fitted values $X\hat{\beta}_\lambda$, the corresponding squared-bias and variance contributions receive an additional factor μ_j . Dividing their sum by n gives the in-design mean prediction error for estimating $X\beta^*$; fresh-response prediction error also adds the irreducible noise term. Coefficient error and prediction error must therefore not be interchanged silently.

Take $\sigma^2 = 1$, $\theta_1 = \theta_2 = 1$, $\mu_1 = 100$, and $\mu_2 = 1$. First complete the four nonbaseline rows. The question marks are intentional: compute them from the general formulas before looking at the reveal on the next page.

μ	λ	s	inverse $\frac{1}{\mu+\lambda}$	squared bias	variance
100	0	1	$\frac{1}{100}$	0	$\frac{1}{100}$
100	1	?	?	?	?
100	9	?	?	?	?
1	0	1	1	0	1
1	1	?	?	?	?
1	9	?	?	?	?

Table 3: Pre-commitment ledger. Baseline rows are supplied; all positive-penalty factors and moment contributions remain hidden.

Commit before the reveal. Circle the row with the largest variance reduction, underline the row with the largest squared bias, and state which value of μ is changed most by a common positive λ .

Reveal and audit. The completed ledger is

μ	λ	s	inverse $\frac{1}{\mu+\lambda}$	squared bias	variance
100	0	1	$\frac{1}{100}$	0	$\frac{1}{100}$
100	1	$\frac{100}{101}$	$\frac{1}{101}$	$\frac{1}{101^2}$	$\frac{100}{101^2}$
100	9	$\frac{100}{109}$	$\frac{1}{109}$	$\frac{9^2}{109^2}$	$\frac{100}{109^2}$
1	0	1	1	0	1
1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
1	9	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{81}{100}$	$\frac{1}{100}$

Table 4: Reveal. The same penalty barely changes the stable direction but sharply reduces variance in the weak direction. The price is displacement of the coefficient mean from its target.

At $\lambda = 1$, the two-direction coefficient mean-squared error is

$$\frac{1 + 100}{101^2} + \frac{1}{4} + \frac{1}{4} \approx 0.510, \quad (48)$$

compared with $\frac{1}{100} + 1 = 1.01$ at $\lambda = 0$. At $\lambda = 9$ it rises to approximately 0.835. This calculation establishes a tradeoff for this declared model; it does not claim that the same λ is optimal for every signal, design, noise level, or prediction distribution.

Definition: Ridge bias-variance tradeoff

Under a declared fixed-design linear model, increasing λ moves the mean ridge coefficient away from the unregularized target while reducing its sensitivity to noise, especially in small-eigenvalue directions. Predictive risk balances the resulting squared bias, variance, and irreducible noise; it need not be monotone in λ .

IN-CLASS CHECK

Prompt: Complete the $\mu = 1$, $\lambda = 4$ row for the same model: give the shrinkage factor, coefficient squared bias, and coefficient variance. Compare its coefficient MSE with $\lambda = 0$ and $\lambda = 9$.

Hint: Use $\theta = 1$ and $\sigma^2 = 1$ only after writing the general formulas.

Answer:

The shrinkage factor is $\frac{5}{17}$. The bias is $-\frac{5}{4}$, so squared bias is $\frac{25}{16}$; variance is $\frac{25}{16}$; their sum is $\frac{25}{8} = 3.125$. In this weak direction, the unregularized MSE is 1, while at $\lambda = 9$ it is $\frac{100}{82} = 1.22$. The moderate penalty $\lambda = 4$ is better than either of those two for this coefficient and this declared signal-noise setting.

IN-CLASS CHECK

Prompt: A classmate says, “larger λ made the training fit worse, so it must have lowered test error.” Identify every logical gap.

Hint: Separate training residuals, coefficient moments, mean prediction error, and fresh noise.

Answer:

A larger penalty generally increases the minimized training residual term; that alone says nothing about test error. The coefficient bias and variance depend on the spectrum, signal coordinates, and noise variance. Prediction error also weights directions through the test-input distribution (or by μ_j for in-design fitted values) and includes fresh noise when predicting responses. None of these quantities is determined by training error alone, and predictive risk need not move monotonically with λ .

Diagnose the failure before choosing the remedy

Suppose four fitted systems all report training MSE 0.20. That equality hides the evidence that matters. The cards below use only quantities already available: repeated-fit behavior, squared bias, fresh-noise scale, and the spectrum of $X^T X$.

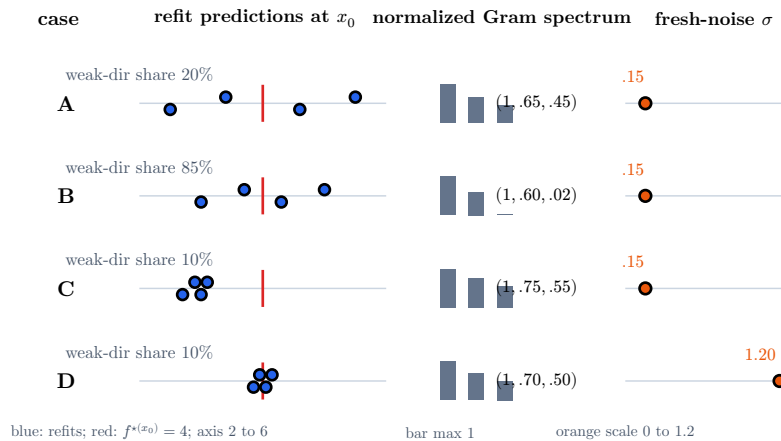


Figure 5: Pre-commitment raw evidence. All cases have training MSE 0.20. The common scales expose refit clouds, eigenvalue bars, weak-direction motion, and fresh-noise marks without naming a diagnosis or remedy.

Commit before the reveal. For every row, write (i) the dominant diagnosis, (ii) a first intervention to defend, and (iii) the later out-of-sample quantity that would test the recommendation. Do not turn the page until all three columns are complete.

Reveal and compare.

case	diagnosis	first intervention	later independent evidence
A	high sample variance	more data or another justified variance control; ridge is not automatic	out-of-sample predictive risk
B	poor conditioning plus variance	ridge is targeted; monitor the bias it adds	out-of-sample predictive risk
C	high systematic bias	richer representation or model, not stronger shrinkage	out-of-sample predictive risk for the revised model
D	high fresh-noise floor	improve measurement or accept the irreducible floor	out-of-sample predictive risk, interpreted against the noise floor

In this course, **diagnosing whether instability calls for regularization** means combining five pieces of evidence: variation across refits, spectral conditioning, systematic prediction bias, fresh-noise scale, and the scientific prediction goal. This is a decision procedure in ordinary language, not a new named mathematical definition.

A useful diagnosis has two layers. First ask whether the observed failure is one that a squared-norm penalty can address. Then specify what an eventual selection procedure must monitor. For Case B, ridge directly attenuates the weak direction, but a penalty that is too large can erase real signal. For Case C, stronger ridge pushes in the wrong direction because systematic bias is already the dominant problem. For Case D, changing the estimator cannot remove fresh observation noise.

IN-CLASS CHECK

Prompt: Work in groups on one case from the matrix. Produce a three-sentence recommendation: (1) diagnose the dominant evidence, (2) defend or reject ridge as the first intervention, and (3) name the prediction quantity that a later selection procedure must estimate.

Hint: A recommendation based only on training MSE is incomplete.

Answer:

Case A: sample variance is high without evidence that it is concentrated in weak spectral directions, so seek more data or another justified variance control; later compare out-of-sample prediction risk. Case B: refit variation aligns with a tiny- μ direction, so ridge is targeted; later compare predictive risk while watching the added bias. Case C: the fitted center is systematically wrong despite stable refits and conditioning, so enrich the representation or model rather than increase shrinkage; later estimate out-of-sample predictive risk for the revised model. Case D: the estimator is stable and centered but the fresh-noise floor is high, so improve measurement if possible or accept the floor; no choice of λ removes irreducible noise.

What SRM does and does not say here

For each radius $B > 0$, define the norm-bounded linear class

$$\mathcal{H}_B = \{x \mapsto \beta^T x : \|\beta\| \leq B\}. \quad (49)$$

These classes are nested as B grows. A norm penalty lets the optimization compare empirical fit against coefficient size, so it can be read as an SRM-style fit-complexity compromise across such nested norm budgets. This is a useful structural analogy.

It is not a statement that adding $\lambda\|\beta\|^2$ literally reduces the VC dimension of an unchanged hypothesis class. A VC-dimension claim would require an explicitly restricted class and the assumptions of a relevant complexity result. The penalty changes the optimization criterion; a class restriction changes the set over which learning occurs. Those statements must not be conflated.

Diagnosis is not parameter selection

The matrix can justify considering ridge and can identify the failure that λ must trade off. It does not choose a numerical value. A legitimate choice requires honest out-of-sample evidence or another justified selection criterion. The final test set remains untouched until all modeling and tuning decisions are complete.

Chapter 5 will introduce training/validation splits, repeated folds, and selection criteria. Those methods are previews here: we do not use one to solve a Chapter 4 calculation, and we do not claim that the data-selected optimum varies continuously with the data.

IN-CLASS CHECK

Prompt: A team tries 40 values of λ and reports the one with the lowest final-test error. What went wrong, and what role should the final test set have?

Hint: Distinguish model fitting, parameter selection, and final evaluation.

Answer:

Repeatedly consulting final-test error makes the test set part of the selection procedure, so its reported minimum is optimistically biased as a final performance estimate. The training data fit candidate models; a separate validation mechanism or justified criterion selects λ ; the untouched final test set is used once for the resulting fixed procedure's final evaluation.

Standard language to carry forward

Term or notation	Meaning in this tutorial	Recall cue
ridge regression	Squared residual fit plus λ times the squared Euclidean coefficient norm.	$\hat{\beta}_\lambda = (X^T X + \lambda I)^{-1} X^T y$
ridge spectral shrinkage	A direction with Gram eigenvalue μ_j retains the fraction $\frac{\mu_j}{\mu_j + \lambda}$ of its OLS coefficient.	small μ_j shrinks most
ridge bias-variance tradeoff	Increasing λ displaces the estimator mean but can reduce noise sensitivity; predictive risk balances both effects.	training fit is not predictive risk

Reconstruct the three items Chapter 5 may use

The next chapter may rely on exactly these three items, in this order:

Available item	Productive recall
empirical risk minimization	Identify the unpenalized training-fit baseline and distinguish it from out-of-sample evaluation.
ridge regression	Derive the registered estimator, read $\frac{\mu_j}{\mu_j + \lambda}$, and explain the fixed-design bias-variance effect.
structural risk minimization	Interpret fit versus controlled class richness without claiming that a penalty automatically changes VC dimension.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct the ridge objective and estimator, state the spectral shrinkage factor, and give one sentence explaining why Chapter 5 still has work to do.

Hint: Keep derivation, interpretation, and selection evidence separate.

Answer:

Ridge minimizes $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ and, for $\lambda > 0$, has $\hat{\beta}_\lambda = (X^T X + \lambda I)^{-1} X^T y$. In a positive Gram eigen-direction, it retains $\frac{\mu_j}{\mu_j + \lambda}$ of the OLS coefficient. Conditioning and fixed-design bias-variance analysis explain what λ changes, but choosing it requires honest out-of-sample evidence or another justified model-selection criterion without tuning on the final test set.

We can now derive a family of regularized estimators and diagnose when that family is relevant. The unresolved question is deliberately statistical: which member predicts best when its training fit is no longer trusted as the answer?

Source note

The historical Week 4 Learning Sheet fixes the official ridge and synthesis scope, but its housing and gene-expression performance numbers are not reused. The objective, normal equations, and positive- λ closed form are reconstructed from Shalev-Shwartz and Ben-David, Section 13.1.1, and Mohri et al., Section 11.3.2, equations (11.12)-(11.14), with the course scaling stated explicitly. The rank-deficient $\lambda \rightarrow 0$ conclusion is derived direction by direction by joining that positive-penalty form to Mohri et al., Section 11.3.1, equation (11.11), which identifies the Moore-Penrose minimum-norm least-squares solution.

The shrinkage interpretation, intercept convention, and distinction between ridge and variable selection follow Claeskens and Hjort, Section 7.6.1, printed pp. 211-213 (PDF pp. 229-231). The fixed-design moment formulas are derived directly from $y = X\beta^* + \varepsilon$ under the displayed covariance assumptions. Shalev-Shwartz and Ben-David, Sections 13.1 and 13.4, support the fit-plus-norm and stability perspectives; the tutorial restricts its SRM claim to nested norm-bounded classes and does not claim a literal VC-dimension reduction for an unchanged class.